

CS602-Computer Graphics

(SUBJECTIVE)

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1. Difference between point and vector?

ANSWER:

Point	Vector
A point position in space.	A Vector no fixed position in space
The only characteristic that distinguishes one point from another is its position.	A Vector has both magnitude and direction.
we draw points as dots	vectors as line segments with arrows.

2. Grouard shading:

ANSWVER:

is a per-vertex color computation. What this means is that the vertex shader must determine a color for each vertex and pass the color as an out variable to the fragment shader. Since this color is passed to the fragment shader as an in varying variable.

3. Phung shading:

ANSWER:

is a per-fragment color computation. The vertex shader provides the normal and position data as out variables to the fragment shader. The fragment shader then interpolates these variables and computes the color.

4. Fractals and Euclidean geometry:

ANSWER:

Fractals are distinct from the simple figures of classical,

Euclidean, geometry—the square, the circle, the sphere, and so forth.

They are capable of describing many irregularly shaped objects or spatially nonuniform phenomena in nature such as coastlines and mountain ranges.

5. Types of fractals:

ANSWER:

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- Fractal Trees
- Fractals in Animal Bodies.
- Fractal Snowflakes.
- Fractal Lightning and Electricity.
- Fractals in Plants and Leaves.

Two basic types of projections are provided for us by OpenGL, along with several corresponding commands for describing the relevant parameters in different ways. One type is the *perspective* projection.

6. OpenGL:

ANSWER

The OpenGL API is a portable API that can be compiled and run on many platforms. OpenGL programs are typically written in C and C++.

7. What is GLUT?

ANSWER:

GLUT (pronounced like the glut in gluttony) is the OpenGL Utility Toolkit, a window system independent toolkit for writing OpenGL programs.

8. Curves

ANSWER:

are regular bends provided in the lines of communication like roads, railways etc. and also in canals to bring about the gradual change of direction. They are also used in the vertical plane at all changes of grade to avoid the abrupt change of grade at the apex.

9. Advantages of curve:

ANSWER:

They provide comfort to the passengers. If there is an abrupt change in the direction nor grade of a highway it will upset the passengers.

10. Define Bezier curve and its formula?

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ANSWER:

A Bezier curve generally follows the shape of the defining polygon.

Formula:

11. Why 24-bit RGB is called true color, define how red green and blue values in it?

ANSWER:

A 24-bit display, of course, uses 24 bits, or 3 bytes per pixel, for color information. This gives 1 byte, or 256 distinct values each, for red, green, and blue. This is generally called *true color*, because 256³ (16.7 million) colors is about as much as your eyes can discern, so more color resolution really isn't necessary, at least for computer monitors.

12. Advantages of plasma display

ANSWER:

- ❖ Can be wall mounted.
- ❖ Less bulky than rear projection television.
- ❖ Virtually less motion blur, so better motion tracking like high refresh rates and faster response time.
- ❖ Far wider viewing angle.
- ❖ More pixels per inch.

13. Disadvantages of plasma display.

ANSWER:

- ❖ Commonly do not come in smaller sizes than 40 inches.
- ❖ Heavier in weight with respect to screen size.
- ❖ A picture is not so clear under bright conditions.
- ❖ Glass screen damage can be permanent and far more difficult to repair.
- ❖ Poor reproduction of black.
- ❖ Radiation is emitted for screen.

14. Write the name of Texture Mapping?

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ANSWER:

1. Perfect Mapping:
2. Affine Mapping
3. Area Subdivision
4. Scan-line Subdivision
5. Parabolic Mapping
6. Hyperbolic Mapping
7. Constant-Z Mapping

15. Write Implicit equation of plane?

ANSWER:

. The implicit equation for a plane is:

$$ax+by+cz+d=0$$

16. write the types of lighting?

There are three basic types of lighting you should layer in a room in order to accomplish this:

- ❖ Ambient
- ❖ general lighting.
- ❖ Accent lighting.
- ❖ Task lighting.

17. Write Specular Reflection with effects?

ANSWER:

Specular reflection from an object produces highlights. Unlike ambient and diffuse reflection, the amount of specular reflection seen by a viewer does depend on the location of the viewpoint - it's brightest along the direct angle of reflection. To see this, imagine looking at a metallic ball outdoors in the sunlight.

18. Effects of OpenGL

ANSWER:

OpenGL allows us to set the effect:

GL_SPECULAR)

GL_SHININESS).

18. Explain Hermite curve?

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A Hermite curve is a spline where every piece is a third degree polynomial defined in Hermite form: that is, by its values and initial derivatives at the end points of the equivalent domain interval.

drawbacks: – Enumerating points on the curve is hard

The hidden surface removal is the procedure used to find which surfaces are not visible from a certain view. A hidden surface removal algorithm is a solution to the visibility issue, which was one of the first key issues in the field of three-dimensional graphics.

19.Explain its important in 3D:

ANSWER:

In 3D computer graphics, hidden surface determination (also known as hidden surface removal (HSR), occlusion culling (OC) or visible surface determination (VSD)) is the process used to determine which surfaces and parts of surfaces are not visible from a certain viewpoint.

20. Explain Modeling Transformations?

ANSWER:

The three OpenGL routines for modeling transformations are `glTranslate*()`, `glRotate*()`, and `glScale*()`. All three commands are equivalent to producing an appropriate translation, rotation, or scaling matrix, and then calling `glMultMatrix*()` with that matrix as the argument. Translate
`void glTranslate{fd}(TYPE x, TYPE y, TYPE z);`

21. Briefly explain Oren—Nayar Diffuse Reflection?

ANSWER:

Though there's been a lot of research on specular reflection models, there's been less research on diffuse reflection models. One of the problems of the standard Lambertian model is that it considers the surface as a smooth diffuse surface. Surfaces that are really rough, like sandpaper, exhibit much more of a backscattering effect, particularly when the light source and the view direction are in the same direction.

22. Two properties of triangle of lambert?

ANSWER:

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Lambert:

Triangles that use Lambertian shading are painted with one solid color instead of using a gradient. Typically, each triangle is lit using that triangle's normal. The resulting object looks very angular and sharp.

23. Advantages of IMPLICIT

ANSWER:

- Better retention over time with less skill loss than explicit.
- More resistance to the effects of psychological stress, disorders and dysfunction.
- Independence of age and IQ.
- Implicit, small individual differences, explicit, large individual differences.

24. Advantages of EXPLICIT

ANSWER:

- ❖ Explicit instruction breaks up learning into smaller parts.
- ❖ This lightens the “cognitive load,” or how much brain resources a student needs to process information. A lighter cognitive load frees up working memory.
- ❖ That's important because learning new skills requires a lot of working memory

25. Advantages of PARAMETER.

ANSWER:

- ❖ Parameters and return values allow students to write programs that are more organized and cleaner.
- ❖ Naming functions helps students write programs that read more like descriptions of what they do, and they also help students reuse code